Juan Francisco Presenza

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Education

2018 oct – 2023 jul — **Ph.D. Engineering, University of Buenos Aires**.
Dissertation title: *Reconfigurable Networks of Unmanned Vehicles.*

2010 mar – 2017 jun — **Electromechanical Engineering, National University of Mar del Plata**.
Integrated 5 year program (3y BSc + 2y MSc) incl. final-year internship and thesis.
Final project title: *Two-Wheeled Inverted Pendulum: A Robotic Prototype for Educational Purposes.*

Research Positions

Postdoctoral Researcher

(Full-time independent postdoctoral research.)

2023 – ***Present*** — **CONICET** (National Scientific and Technical Research Council, Argentina).
Host institution: University of Buenos Aires .
Instituto de Tecnologías y Ciencias de la Ingeniería "Hilario Fernández Long" (Institute of Engineering Technology and Sciences).
Research focus: *Graph rigidity theory and its applications to multi-robot control.*

Doctoral Researcher

(Full-time doctoral research leading to the award of the Ph.D.)

2018 – 2023 — **University of Buenos Aires** (Argentina).
Laboratorio de Automática y Robótica (Automation and Robotics Laboratory).
Research focus: *Scalable control methods for networks of unmanned vehicles.*

Research Publications

*(*Main Contributor.)*

Journal Articles

1. J. I. Alvarez-Hamelin, J. I. Giribet, I. Mas, and **J. F. Presenza***, "Extremal properties and bounds for the generalized algebraic connectivity of graphs in euclidean spaces," *Linear Algebra and its Applications*, vol. 727, pp. 24–36, 2025, ISSN: 0024-3795. DOI: https://doi.org/10.1016/j.laa.2025.07.028.
2. **J. F. Presenza***, I. Mas, J. Ignacio Alvarez-Hamelin, and J. I. Giribet, "Subframework-based bearing rigidity maintenance control in multirobot networks," *IEEE Control Systems Letters*, vol. 9, pp. 1249–1254, 2025, Presented at the 2025 Conference on Decision and Control. DOI: 10.1109/LCSYS.2025.3580768.
3. P. Moreno, **J. F. Presenza**, I. Mas, and J. I. Giribet, "Aerial multi-camera robotic jib crane," *IEEE Robotics and Automation Letters*, vol. 6, no. 2, pp. 4103–4108, 2021. DOI: 10.1109/LRA.2021.3065299.
4. C. Pose, **F. Presenza**, I. Mas, and J. I. Giribet, "Trajectory following with a mav under rotor fault conditions," *Unmanned Systems*, vol. 08, no. 04, pp. 263–268, 2020. DOI: 10.1142/S2301385020500193.

Conference Proceedings

1. E. Pecker Marcosig, **J. F. Presenza**, I. Mas, J. I. Alvarez-Hamelin, J. I. Giribet, and R. Castro, "Simulation of decentralized coordination strategies for networked multi-robot systems with emergent behavior-devs," in *Proceedings of the 2025 Winter Simulation Conference*, 2025. DOI: `10.1109/WSC68292.2025.11338929`.
2. **J. F. Presenza***, J. I. Alvarez-Hamelin, I. Mas, and J. I. Giribet, "Subframework-based rigidity control in multirobot networks," in *2022 American Control Conference (ACC)*, 2022, pp. 3431–3436. DOI: `10.23919/ACC53348.2022.9867693`.

Books Chapters

1. **J. F. Presenza***, E. Pecker Marcosig, I. Mas, J. I. Álvarez-Hamelin, and J. I. Giribet, *Scalable Rigidity Maintenance Strategies for Networked Multirobot Systems: Overcoming Decentralized Coordination Challenges*, C. Szabo, V. Semeshenko, R. Castro, and S. Mittal, Eds. John Wiley & Sons, 2026, ch. 14, Forthcoming, ISBN: 978-1-394-32627-3.

Refereed Conference Papers

1. **J. F. Presenza***, I. Mas, J. I. Giribet, and J. I. Alvarez-Hamelin, "Descomposición óptima en subframeworks con aplicaciones al control de redes multirobot," XII Jornadas Argentinas de Robótica, 2024.
2. **J. F. Presenza***, J. I. Giribet, J. I. Alvarez-Hamelin, and I. Mas, "Control de rigidez por subframeworks en sistemas multirobot," XI Jornadas Argentinas de Robótica, 2022.
3. E. Pecker Marcosig, **J. F. Presenza**, J. I. Giribet, and I. Mas, "Sistema de control para un robot aerodeslizante," X Jornadas Argentinas de Robótica, 2019.
4. **J. F. Presenza***, P. Claudio, I. Mas, and J. I. Giribet, "Validación de un sistema de posicionamiento ultrasónico para interiores," X Jornadas Argentinas de Robótica, 2019.
5. **J. F. Presenza***, E. M. Gelos, S. A. González, and J. R. Fischer, "Plataforma móvil basada en un péndulo invertido de dos ruedas," XVII Reunión de Trabajo sobre Procesamiento de la Información y Control, 2017.

Submitted Manuscripts

1. **J. F. Presenza***, L. J. Colombo, J. I. Giribet, and I. Mas, *Angle-based localization and rigidity maintenance control for multi-robot networks*, under revision. Preprint available at https://arxiv.org/abs/2604.11754, 2026.
2. **J. F. Presenza***, L. J. Colombo, I. Mas, and J. I. Giribet, *Multi-robot bearing-only pose estimation via angle rigidity*, under revision. Preprint available at https://arxiv.org/abs/2606.03931, 2026.

Teaching Positions

Lecturer (part-time) — Instructor of Record

(Full responsibility for course design, lectures, assessment, and grading. ∼ 60 contact hours per course edition.)

2025 - *Present* — **Harmonic Analysis** (Undegraduate - Engineering).
Department of Engineering, University of San Andrés, Argentina.
1 semester-long course.

Teaching Positions (continued)

2024 – *Present* — **Discrete Mathematics** (Undegraduate - Engineering).
Department of Engineering, University of San Andrés, Argentina.
3 semester-long courses.

Lecturer (part-time) — Tutorial Instructor

(Regular tutorial lecturing and collaboration on course design, assessment, and grading. $\sim$ 60 *contact hours per course edition.)*

2019 – 2025 — **Linear Algebra** (Undegraduate - Engineering).
Department of Mathematics, University of Buenos Aires, Argentina.
13 semester-long courses.

2022 – 2024 — **Discrete Mathematics** (Undegraduate - Engineering).
Department of Engineering, University of San Andrés, Argentina.
4 semester-long courses.

2022 — **Linear Algebra** (Undegraduate - Engineering).
Department of Engineering, University of San Andrés, Argentina.
1 semester-long course.

Teaching Assistant (Part-time)

(Regular tutorial lecturing and collaboration on course content. $\sim$ 60 *contact hours per course edition.)*

2013 – 2016 — **Engineering Mathematics** (Undegraduate - Engineering).
Department of Mathematics, National University of Mar del Plata, Argentina.
8 semester-long courses.

Funded Research Projects and Fellowships

Research Projects

2023 - **Present** — **University of Buenos Aires** (Internal research grant).
Project title: *Dynamics of Information Flow on the Internet and Social Networks from a Complex Systems Perspective.*
Role: Postdoctoral researcher. P.I.: J. Ignacio Alvarez-Hamelin.

Fellowships

2023 - **Present** — **Postdoctoral Fellowship**, CONICET.
Individually awarded competitive national fellowship.

2018 - 2023 — **Doctoral Fellowship**, University of Buenos Aires.
Individually awarded competitive national fellowship.

Supervision

Undergraduate thesis

Ongoing — **Manuel Nuñez, co-supervized with Dr. Ezequiel Pecker-Marcosig**
Engineering degree: 6 year program, approximately equivalent to BSc + MSc.
Thesis title: *DEVS and ROS2 Integrated Embedded Simulation for Multi-Robot Formations*

2024 — **Eng. Luana Quirolo, co-supervized with Dr. Juan Giribet**.
Engineering degree: 6 year program, approximately equivalent to BSc + MSc.
Thesis title: *Robot Velocity Estimation with Optical Flow Sensors*

Research and Educational Software

- **Rigidity-based Multi-Robot Coordination**.
  `https://github.com/fpresenza/uv_network`
- **Event-based Simulation Library for Dynamical Multi-Agent Systems**.
  `https://github.com/fpresenza/EBDEVS-multiagent`
- **Applied Mathematics for Educational Purposes**.
  `https://github.com/fpresenza/appliedmath`
- **Control and Estimation of an Omnidirectional Mobile Robot with Arduino and ROS**.
  `https://github.com/fpresenza/omniduino`
- **Rust-based Cryptographic Tools**.
  `https://github.com/fpresenza/sparkling_water_bootcamp`

Skills

Languages	Spanish: Native proficiency.
	English: Proficient (C1/C2 level, CEFR). (5 year coursework and professional use).
	French: Elementary (A2 level, CEFR) (1 year coursework).
Programming	Python, C/C++, Rust, R, MATLAB, Git, LaTeX.
Scientific Computing	SageMath, NetworkX, Graph-tool, PowerDEVS, PyPDEVS
Robotics Tools	ROS/ROS2, Gazebo, PX4, Raspberry Pi, Arduino.

Awards and Honors

2023 — **Ph.D. Thesis evaluated as "sobresaliente" (outstanding)** by the examination committee, University of Buenos Aires.

2017 — **Graduated with Diploma of Honor (average grade: 8.67/10)**, National University of Mar del Plata.

Certification

Languages

2013 — **French Language – 1st Year Completed (approx. CEFR level A2)**. Laboratorio de Idiomas, National University of Mar del Plata.

2012 — **English Language – 5th Year Completed (approx. CEFR level C1/C2)**. Laboratorio de Idiomas, National University of Mar del Plata.

Advanced Coursework

2024 — **Applied Cryptography**, Lambda Class (`https://lambdaclass.com/`).

2021 — **Convex Optimization**, University of Buenos Aires.

2020 — **Statistical Learning**, University of Buenos Aires.

2019 — **Robust Control Theory**, University of Buenos Aires.

- **Integrated Navigation Systems**, University of Buenos Aires.
- **Mobile Robotics: A Probabilistic Approach**, University of Buenos Aires.

Additional Professional Experience

2017 jul – 2018 sep **Junior Professional Program**, Trenes Argentinos (National railway operator).